🚀 Credit Risk Analysis

SQL + EDA + Machine Learning + Business Insights

📌 Project Overview

Developed a **credit risk prediction system** to assess whether a loan applicant is likely to **default or repay**, using real-world financial data.

The project combines:

- **SQL** → Data extraction & transformation
- **EDA** → Pattern discovery
- **Machine Learning** → Risk prediction
- **Business Insights** → Decision-making support

🎯 Problem Statement

Financial institutions face **high losses due to loan defaults**.

This project aims to:

- Identify **high-risk applicants**
- Reduce **financial losses**
- Improve **loan approval decisions**

🛠️ Tech Stack

- **Python** (Pandas, NumPy, Scikit-learn)
- **SQL (PostgreSQL)**
- **Matplotlib / Seaborn**
- **Jupyter Notebook**

📊 Dataset

- Loan applicant data including:
  - Income
  - Credit history
  - Loan amount
  - Employment status

    - Repayment status (target variable)

🔍 **Step 1: Data Extraction using SQL**

- Queried structured financial data from PostgreSQL
- Performed:
    - Filtering invalid entries
    - Joining multiple tables
    - Aggregations (avg income, loan ratios)

👉 Example:

```
SELECT applicant_income, loan_amount, credit_history
FROM loan_data
WHERE loan_status IS NOT NULL;
```

📈 **Step 2: Exploratory Data Analysis (EDA)**

- Handled missing values
- Analyzed distributions & correlations

Key insights:

- Applicants with **low credit history score → high default rate**
- **Higher loan-to-income ratio → risky loans**
- Self-employed applicants had **slightly higher default probability**

🤖 **Step 3: Machine Learning Model**

- Models used:
    - Logistic Regression
    - Decision Tree
    - Random Forest (best performance)
- Tasks:
    - Feature encoding
    - Train-test split
    - Model evaluation

👉 Metrics:

- Accuracy
- Precision & Recall
- Confusion Matrix

📊 **Step 4: Results**

- Achieved ~**X% accuracy** (replace with yours)
- Random Forest performed best in:
  - Handling non-linear patterns
  - Reducing false approvals

💡 **Business Insights**

- Reject or review applications with:
  - Low credit history
  - High loan-to-income ratio
- Prioritize applicants with:
  - Stable income
  - Strong repayment history

👉 Impact:

- Helps banks **minimize risk**
- Improves **loan approval efficiency**

🌐 **Future Improvements**

- Integrate **real-time loan APIs**
- Deploy as a **web dashboard**
- Add **credit score explanation (Explainable AI)**

📁 **GitHub Structure (VERY IMPORTANT for you)**

```
Credit-Risk-Analysis/
├── data/
├── notebooks/
├── sql/
```

```
│── models/
│── app/ (optional dashboard)
│── README.md
│── requirements.txt
```

🧠 **Resume Bullet (Use THIS exactly)**

- Built a **credit risk prediction system** using SQL, EDA, and machine learning, achieving improved loan default classification and generating actionable business insights for risk reduction.

🔥 **If you want to level this up (highly recommended)**

Add ONE of these:

- Streamlit dashboard (huge boost)
- Explainable AI (SHAP values)
- API deployment (Flask/FastAPI)

If you want, I can next:

- Turn this into a **perfect GitHub README (with badges + visuals)**
- Give you **full working code (EDA + ML pipeline)**
- Or help you **deploy this as a dashboard (major resume boost)**